

Experiment – 1

Aim : - Design web pages for your college containing a description of the courses, departments, faculties, library etc, use href, list tags .

Thoery :

HTML (HyperText Markup Language) is the standard language used to create web pages. It provides various tags to structure content such as headings, paragraphs, images, and links. The `<a>` tag with the href attribute is used to create hyperlinks that connect different web pages, enabling easy navigation. List tags like `<ul>` (unordered list), `<ol>` (ordered list), and `<li>` (list items) help organize information in a readable format. This experiment focuses on building a structured informational website using basic HTML concepts and navigation techniques.

Experiment – 2

Aim : - Write html code to develop a webpage having two frames that divide the webpage into two equal rows and then divide the row into equal columns fill each frame with a different background color.

Theory : -

Frames are used in HTML to divide a browser window into multiple sections, where each section can display a different document. The `<frameset>` tag replaces the `<body>` tag and is used to define rows and columns. Each frame is defined using the `<frame>` tag. Background colors can be applied using attributes or CSS to distinguish different frames. Although frames are now deprecated in modern HTML, this experiment helps in understanding basic layout concepts and how content can be visually separated.

Experiment – 3

Aim : - Design a web page of your home town with an attractive background color, text color, an Image, font etc. (use internal CSS).

Theory :

CSS (Cascading Style Sheets) is used to control the presentation and layout of web pages. Internal CSS is written within the <style> tag inside the <head> section of the HTML document. It allows styling of multiple elements within a single page, including colors, fonts, spacing, and alignment. By combining HTML structure with CSS styling, visually attractive and user-friendly web pages can be created. This experiment emphasizes enhancing the appearance of content using internal styling techniques.

Experiment – 4

Aim : - Use External, Internal, and Inline CSS to format college web page that you created.

Thoery : -

CSS can be implemented in three different ways:

- **Inline CSS:** Applied directly within an HTML element using the style attribute.
- **Internal CSS:** Defined inside the <style> tag within the <head> section.
- **External CSS:** Written in a separate .css file and linked using the <link> tag.

Each method has its own advantages. External CSS is most efficient for large websites, while inline CSS is useful for quick styling. This experiment demonstrates how different CSS approaches can be used together to improve layout, consistency, and design.

Experiment – 5

Aim : - Create HTML Page with JavaScript which takes Integer number as input and tells whether the number is ODD or EVEN.

Theory : -

JavaScript is a client-side scripting language used to add interactivity to web pages. It allows user input to be processed instantly without server communication. In this experiment, conditional statements (if-else) are used to check whether a number is divisible by 2. If the remainder is zero, the number is even; otherwise, it is odd. This experiment demonstrates basic programming logic, user interaction, and event handling in web development.

Experiment – 6

Aim : - Create HTML Page that contains form with fields Name, Email, Mobile No, Gender , Favourite Colour and a button now write a JavaScript code to combine and display the information in textbox when the button is clicked and implement validation.

Theory : -

HTML forms are used to collect user input through various elements like text boxes, radio buttons, checkboxes, and dropdown menus. JavaScript is used for client-side validation to ensure that the entered data meets required criteria (e.g., valid email format, non-empty fields). It also allows dynamic display of user data without refreshing the page. This experiment highlights the importance of form validation and enhances user experience by preventing incorrect data submission.

Experiment – 7

Aim : - Create XML file to store student information like Enrolment Number, Name Mobile Number , Email Id.

Thoery : -

XML (Extensible Markup Language) is a flexible text-based format used to store and transport data. It uses custom-defined tags to represent structured information. XML is platform-independent and widely used for data exchange between systems. Each piece of data is enclosed within meaningful tags, making it easy to understand and process. This experiment demonstrates how structured data can be organized and stored using XML.

Experiment – 8

Aim : - Write a php script to read data from txt file and display it in html table (the file contains info in format Name: Password: Email).

Thoery : -

PHP is a server-side scripting language used to develop dynamic web applications. It provides built-in functions such as `fopen()`, `fgets()`, and `fclose()` for file handling. Data stored in a text file can be read line by line and displayed dynamically on a web page. By combining PHP with HTML, the retrieved data can be formatted into tables for better readability. This experiment demonstrates file operations and dynamic content generation.

Experiment – 9

Aim : - Write a PHP Script for login authentication. Design an html form which takes username and password from user and validate against stored username and password in file.

Thoery : -

Login authentication is an important aspect of web security. In this experiment, user credentials (username and password) are stored in a file. PHP reads the stored data and compares it with the user input received from an HTML form. If the credentials match, access is granted; otherwise, an error message is displayed. This introduces basic concepts of authentication, validation, and secure data handling.

Experiment – 10

Aim : - Write PHP Script for storing and retrieving user information from MySQL table -

- 1) a. Design A HTML page which takes Name, Address, Email and Mobile No. From user (register.php)
- 2) b. Store this data in Mysql database / text file.
- 3) c. Next page display all user in html table using PHP (display.php)

Thoery : -

MySQL is a relational database management system used to store structured data. PHP can connect to MySQL using functions like `mysqli_connect()` or PDO (PHP Data Objects). Data collected through HTML forms is inserted into database tables using SQL queries. It can also be retrieved and displayed dynamically in an HTML table. This experiment demonstrates database connectivity, CRUD operations (Create, Read, Update, Delete), and integration of frontend and backend technologies.

Experiment – 11

Aim : Design a responsive website for a fictional event, like a tech conference or a cultural fest. Use media queries in CSS to ensure the website is viewable on devices of various sizes. Test the website on different devices and document the responsiveness.

Theory :

Responsive Web Design (RWD) is an approach used in web development to create websites that automatically adjust their layout, content, and appearance according to the screen size and device type. With the increasing use of smartphones, tablets, and desktops, it is essential to ensure that websites are accessible and user-friendly across all devices.

The core idea behind responsive design is to use fluid layouts, flexible images, and CSS media queries. Instead of using fixed widths (like pixels), developers use relative units such as percentages, em, or rem so that elements resize proportionally.

Key concepts:

- Media Queries: Apply CSS based on screen size
- Flexible Layouts: Using percentages instead of fixed widths
- Viewport Meta Tag: Controls layout on mobile

Experiment – 12

Aim : Create a single-page application using a modern JavaScript framework or library (React, Angular, or Vue.js). The application should display different content when users navigate to different sections, without reloading the page.

Theory :

A Single Page Application (SPA) is a type of web application that loads a single HTML page and dynamically updates its content without reloading the entire page. Modern SPAs are commonly built using frameworks like React, Angular, or Vue.js.

Unlike traditional websites where each user action (like clicking a link) loads a new page from the server, SPAs use JavaScript to manipulate the DOM and update only specific parts of the page. This results in faster performance and smoother user experience.

Key Concepts

- Components: Reusable pieces of UI (e.g., navbar, footer).
- State Management: Keeps track of data changes.
- Routing: Enables navigation without page reload.
- Virtual DOM (in React): Improves performance by updating only changed parts.

Experiment – 13

Aim : Develop a web application that interacts with an external RESTful API. Fetch data from a public API (like a weather service or a public data set) and display the results in a user-friendly format on your webpage.

Theory :

A RESTful API (Representational State Transfer API) is a web service that allows communication between a client (browser) and a server using HTTP protocols. It follows REST principles, which are lightweight and widely used in modern web applications.

In this experiment, the web application interacts with an external API to fetch data and display it dynamically on a webpage.

Key Concepts of REST

- Client-Server Architecture: Separation between frontend and backend
- Statelessness: Each request contains all required information
- Uniform Interface: Standard HTTP methods are used

HTTP Methods

- GET → Retrieve data
- POST → Send data
- PUT → Update data
- DELETE → Remove data

Data Format

Most APIs return data in JSON (JavaScript Object Notation) format, which is easy to parse and use in JavaScript.